

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)

Second Semester M. Tech. (Construction Technology) (CBCS) Degree Examinations

May - June 2024

22CST213 – DESIGN CONCEPTS OF SUBSTRUCTURES

Duration: 3 Hours

Max. Marks:100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

❖ Assume all the missing data suitably as per Indian standards

Unit – I

- | | Marks | BT* | CO* | PO* | | | | | | | | | |
|--|--------------|------------|------------|------------|-----|-----|---------|-----|-----|--|--|--|--|
| 1. a) (i) Explain in brief the significance of site investigation report. | 4 | | | | | | | | | | | | |
| (ii) Illustrate the types of foundations as per IS: 1904 - 1986 | 6 | L*2 | 1 | 1 | | | | | | | | | |
| b) Square footing 1.2 m x 1.2 m rests at a depth of 1 m in a saturated clay layer 4 m deep. The clay is normally consolidated, having an unconfined compressor strength of 40 kN/m ² . The soil has a liquid limit of 30% $\gamma_{sat}=17.8$ kN/m ³ $w=28\%$ and $G=2.68$ Determine the load which the footing can carry safely with the factor of safety of 3 against shear Also determine the settlement if the footing is loaded with the safe load. Use Terzaghi's analysis for bearing capacity. | 10 | L2 | 2 | 2 | | | | | | | | | |
| 2. a) Proportion a strap footing for the following data:-
Allowable pressures: 150kN/m ² for D.L + reduced L.L
225kN/m ² for D.L + L.L; Column loads are as follows: | | | | | | | | | | | | | |
| <table border="1"> <thead> <tr> <th></th> <th>Column A</th> <th>Column B</th> </tr> </thead> <tbody> <tr> <td>D.L(KN)</td> <td>500</td> <td>600</td> </tr> <tr> <td>L.L(KN)</td> <td>450</td> <td>800</td> </tr> </tbody> </table> | | Column A | Column B | D.L(KN) | 500 | 600 | L.L(KN) | 450 | 800 | | | | |
| | Column A | Column B | | | | | | | | | | | |
| D.L(KN) | 500 | 600 | | | | | | | | | | | |
| L.L(KN) | 450 | 800 | | | | | | | | | | | |
| • Proportion the footing for uniform pressure under D.L + reduced L.L. The c/c distance of the columns = 5.4m. The projection beyond column A not greater than 0.5m. | 10 | L3 | 1 | 2 | | | | | | | | | |
| b) What are combined footings? Under what circumstances one can recommend them? Explain the different types of combined footings with illustrations. | 10 | L2 | 2 | 1 | | | | | | | | | |
| 3. a) A trapezoidal footing is to be produced to support two square columns of 30 cm and 50c sides respectively. Columns are 6 apart and the SBC of the soil is 400kN/m ² . The bigger column carries 5000kN and the smaller 3000kN. Design a suitable size of the footing so that it does not extend beyond the faces of the columns. | 10 | L3 | 1 | 1 | | | | | | | | | |
| b) What are the different type of boring? Explain any two types with neat sketches. | 10 | L1 | 2 | 3 | | | | | | | | | |

Unit – II

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|---|----|----|---|---|
| 4. a) Enumerate any four situations suitable for providing pile foundation? How piles are classified based on function? Discuss in detail. | 10 | L2 | 3 | 1 |
| b) A nine pile group arranged in a square pattern is used as a foundation for a column in sand ($\phi' = 32^\circ$). Piles 300mm in diameter and 10m in length, are placed at a spacing of 900mm in each direction. Calculate the ultimate load capacity of the pile group. Assume the unit weight of soil as 18kN/m ³ . Use Berezantsev recommendation for bearing capacity factor, N_q . | 10 | L3 | 4 | 1 |
| 5. a) A precast concrete pile of size 35cm×35cm is driven by a single acting steam hammer. Estimate the allowable load using.
i) Engineering News Record Formula with FOS = 6.
ii) Modified Hiley's Formula using FOS =4 (IS code method)
Use the following details: Maximum rated energy =3500 KN-cm; Weight of hammer =35kN Length of pile =15m; Efficiency of hammer =0.8; Co- efficient of restitution =0.5; Weight of pile cap =3kN. No. Of blows for last 25.4mm =6; Modulus of elasticity of concrete = 2×10^7 kN/m ² . Take the weight of the pile as 73.50kN and assume any other data if required. | 10 | L3 | 3 | 1 |

P.T.O

- b) Explain with the help of neat sketches the following classification of deep foundation a) Pile foundation b) Caissons.

10 L2 4 .3

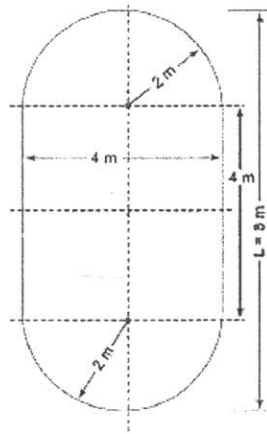
6. a) Write a note on (i) Negative skin friction (ii) Uplift capacity of a pile.
b) A 300mm diameter concrete pile, 10 m long was driven by a McKiernan and Terry double acting hammer (rated energy 16.62kJ, total mass 2200kg). The driving was done with a short dolly and cushion 2.50m. The average penetration recorded in the last five blows was 3.00mm/blow. Calculate the safe pile load.

10 L2 3 1

10 L3 4 2

Unit – III

7. a) Find the maximum base pressure and lateral load per unit length of the well base, for a well shown in Fig 7a. The following data are given Effective downward force on well, including self weight = 1000t. Height of point of application of horizontal force above the scour line = 9m. depth of well below the scour line = 9m Submerged unit weight of soil = 0.78t/m^3 . Angle of internal friction = 30° Coefficient of vertical subgrade reaction = $25 \times 10^5 \text{ kg/cm}^2$. Poisson's ratio = 0.5, Safe bearing capacity of soil at the base of the well = 4.5kg/cm^2 . Horizontal deformation of the well cap = 2cm ($=p_1$). Assume that the well rotates about its base.



- b) With a neat sketch explain the components of a well foundation.

10 L3 5 1

10 L2 5 2

8. a) Describe the various forces acting on well foundations.
b) Determine the outside diameter of an open caisson to be sunk through 40 m of sand and water to bed rock if the allowable bearing capacity is 2000kN/m^2 . The caisson receives a load of 50MN from the superstructure. The mantle friction is 30kN/m^2 . Test the feasibility of sinking. Also calculate the thickness of the seal.

10 L2 5 1

10 L3 5 3

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
